



Digital Image Processing

Prof. Zohreh Azimifar

Final Project

**From Foundations to State-of-the-Art: A Comprehensive
Synthesis**

Submission Format: Phased Presentations to TAs + Final
Comprehensive Slide Deck (PDF)

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1. Project Overview & Philosophy

Throughout this course, you presented foundational papers and concepts in digital image processing (DIP). While the classical methods established the theoretical groundwork of the field, modern computer vision and image processing have evolved rapidly, driven by deep learning and massive compute capabilities.

The objective of this final project is to track the evolution of your specific topic from its classical DIP origins to its current State-of-the-Art (SOTA). **A core requirement of this project is demonstrating the clear, evolutionary path of your topic.** For example, if your topic is edge detection, you must map the journey from classical approaches (e.g., the Canny Edge Detector), to early deep learning milestones (e.g., Holistically-Nested Edge Detection (HED)), all the way to current SOTA models (e.g., Transformer-based edge detectors).

To truly demonstrate your mastery of a complex topic, you will act as domain experts. You will conduct an exhaustive literature review, connect theoretical evolution to modern engineering practices (e.g., GitHub, Hugging Face), and synthesize your findings into a massive, highly detailed presentation deck containing enough depth to fuel a 2 to 4-hour masterclass.

2. The 3-Stage Project Pipeline & TA Checkpoints

This project is divided into three distinct stages. For each stage, your group is required to present your progress to the class Teaching Assistants (TAs). These checkpoints ensure your research is on track, your sources are high-quality, and your final synthesis is cohesive.

Stage 1: Exhaustive Literature Review & Timeline Construction

Your first task is to map the academic landscape of your assigned topic. You must trace how the problem was solved in the past using classical DIP, the bottlenecks that arose, and how modern researchers are solving it today.

- **Source Quota:** You must curate, read, and analyze a minimum of **10 distinct, high-quality sources**.
- **Source Distribution Rules:**
 - **Category A (Minimum 70%):** Peer-reviewed academic papers. These must be sourced from top-tier journals and conferences (e.g., IEEE TPAMI, CVPR, ICCV, ECCV, NeurIPS, MICCAI). You must include a mix of older milestone papers and recent SOTA publications (2022–present).
 - **Category B (Maximum 30%):** Reputable auxiliary materials. This includes textbook chapters, recognized academic lectures (e.g., Stanford CS231n, MIT OpenCourseWare), or highly technical developer blogs from industry leaders (e.g., Google Research, Meta AI).

- **The Goal:** Do not merely summarize the papers. You must establish a chronological and thematic connection between them, **explicitly highlighting the path from classical algorithms to modern SOTA**. How did the field transition from spatial filtering to CNNs, and eventually to Vision Transformers?

TA Presentation 1: Literature Review Pitch (Deadline: 11 Aug 2026)

- **Deliverable:** A short slide deck detailing your selected topic, your exact list of 10+ sources, and a visual timeline mapping the evolution of the algorithms (e.g., *Classical Math* → *CNN* → *Transformer*).
- **Objective:** The TAs will evaluate the rigor of your chosen papers, ensure you meet the 70/30 split, and approve your research direction.

Stage 2: Practical Synthesis & Modern Engineering

Modern image processing is not just about mathematical theory; it is about software engineering, framework optimization, and open-source deployment. In this stage, you must bridge the gap between the academic papers from Stage 1 and real-world implementation.

- **Identifying the Modern Standard:** What is the current industry and academic standard for your topic? You must clearly contrast how these modern architectures solve the problems that classical DIP algorithms struggled with.
- **Codebase & Repository Analysis:** You must find and analyze practical, open-source implementations of the modern approaches.
 - Explore platforms like **GitHub, Hugging Face, and PapersWithCode**.
 - You are not required to code a new architecture from scratch, but you *are* required to perform a "code deep-dive."
 - You must explain how these modern models are structured, what frameworks they use (PyTorch/TensorFlow), how their data pipelines work, and how pre-trained weights are utilized today compared to older, manual feature-extraction methods.

TA Presentation 2: Practical Engineering & Code Analysis (Deadline: 21 Aug 2026)

- **Deliverable:** A presentation mapping the theories from your papers to specific GitHub/Hugging Face repositories.
- **Objective:** You will walk the TAs through the architecture of a modern repository related to your topic. Show them you understand the code structure, the modern loss functions being used, and the practical deployment realities of the SOTA models.

Stage 3: The "Deep-Understanding" Unified Slide Deck

For your final deliverable, you will combine your historical research (Stage 1) and your practical code analysis (Stage 2) into a single, cohesive, and highly detailed master presentation.

- **Scope & Depth (2 to 4 Hours of Content):** This is not a standard 15-minute presentation. The depth, density, and flow of the information in this slide deck must be equivalent to a 2 to 4-hour advanced lecture.
- **Purpose:** While you will not actually present for 4 hours, the deck itself serves as a standalone educational artifact. It is designed to prove your absolute, bottom-to-top mastery of the subject. If an incoming graduate student wanted to learn everything about your topic—from its classical mathematical origins to downloading its modern weights on Hugging Face—your slide deck should be their ultimate guide.
- **Cohesion:** The deck must flow logically. It cannot read like 10 disconnected paper summaries. It must tell the complete evolutionary story of your domain, constantly referencing back to how the field has evolved.

TA Presentation 3: Final Project Defense (Deadline: 31 Aug 2026)

- **Deliverable:** The final PDF slide deck submitted for grading.
- **Objective:** Since a 4-hour presentation is impossible due to time constraints, you will present a highly condensed executive summary of your massive deck to the TAs. You will be questioned on various parts of your full slide deck to verify your deep understanding of the material you curated.

3. Guidelines for Structuring the Final Slide Deck

To ensure your final presentation meets the extensive "masterclass" requirement without relying on filler content, we recommend the following structural flow for your slides:

1. **Introduction & Problem Definition:** Clearly define the image processing task. Why is it important? What are the inherent mathematical or visual challenges?
2. **The Classical DIP Foundation:** A review of the traditional image processing algorithms (e.g., Canny Edge Detector) that you presented in class. What was the original mathematical solution? What were its limitations (e.g., reliance on hard-coded thresholds, inability to handle complex noise)?
3. **The Deep Learning Transition (Early Milestones):** The bridge between classical and modern. Explain the early structural shifts (e.g., moving from the Canny algorithm to CNN-based approaches like Holistically-Nested Edge Detection).
4. **The Modern State-of-the-Art:** Detailed explanation of the absolute best methods used today (e.g., Transformer-based models). Include modern network architectures, attention mechanisms, and advanced training paradigms.
5. **Practical Deep-Dive (Engineering Reality):** Screenshots, snippets, and architectural diagrams sourced from modern GitHub and Hugging Face repositories. Walk the reader through how these SOTA models are actually coded, trained, and deployed today.
6. **Comparative Analysis:** Visual and quantitative comparisons (PSNR, SSIM, IoU) showcasing the massive leap in performance from the classical DIP methods to the modern SOTA models.
7. **Future Directions & Conclusion:** What are the current unsolved problems in this specific domain?

4. Evaluation & Grading Criteria

Your final grade will be determined by:

1. **Rigor of Research (Stage 1):** The quality, relevance, and difficulty of the 10+ selected sources, and your ability to accurately trace the classical-to-modern evolutionary path.
2. **Practical Integration (Stage 2):** Your ability to analyze and explain complex, modern open-source repositories (GitHub/Hugging Face).
3. **Depth & Cohesion (Stage 3):** Whether your final slide deck successfully achieves the required 2-to-4-hour depth of a unified, comprehensive masterclass.
4. **TA Defenses:** Your performance, clarity, and ability to successfully answer technical questions during the three mandatory TA checkpoints.